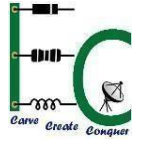




The National Institute of Engineering, Mysuru

Department of Electronics and Communication Engineering



Virtual Lab Report

Course: Linear Integrated Circuits

Course Code: BEC405D

Submitted by

Bhoomika M

4NI24EC028

4TH SEM, ECE,"C" SECTION

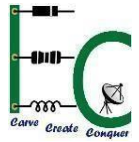
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LAB SHEET 7

Name: Bhoomika M

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Date: 26-04-2026

PRECISION RECTIFIER

Aim: To design, build and test Precision Rectifiers for given specifications.

1. To understand the concept of super diode.
2. To implement inverting and non-inverting half wave rectifier.
3. To implement inverting and non-inverting full wave rectifier.
4. Plot input and output waveforms.

Components/Parts required: μ A741 IC (Op-Amp), resistors, diodes (IN4007), signal generator, output meter, DC power supply.

Prelab:

Theory:

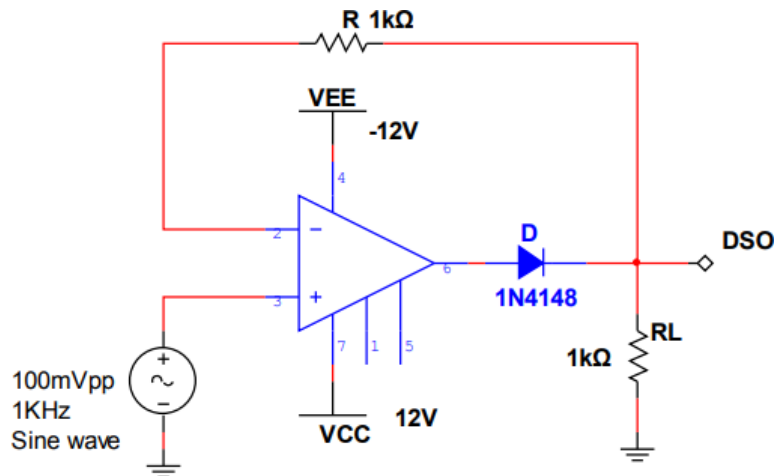
A rectifier is a DC voltage source. It has finite output impedance in series with a DC voltage source. Therefore a rectifier can be modelled as a constant voltage source in series with a resistance. The rectification becomes a crucial problem when we have to rectify millivolt signal at high frequencies. This is because the conventional diodes require at least 0.2 V (Ge) or 0.7 V (Si) across them for a forward bias and hence a millivolt signal cannot forward bias a conventional diode. Also the switching speed of conventional diode is very low. If we use a diode in the forward path of op amp cut in voltage of the diode will be divided by the large open loop gain of op amp, thus making the diode as ideal one i.e. a diode with zero cut in voltage. When we use a conventional diode in the feedback path of an op amp, we can rectify millivolt signals at high frequencies. This set up is then called as Precision rectifiers. The combination of diode and op-amp in precision rectifier circuit is called as Superdiode. In case of conventional rectifier, output peak will be less than the input peak. This is due to loss of voltage across the diode due to finite cut in voltage. But in the case of precision rectifier diodes, it will have zero cut in voltage and hence output peak will be same as input peak. Precision rectifiers are classified as Half wave precision rectifier and Full wave precision rectifier according to the operation of

the circuit.

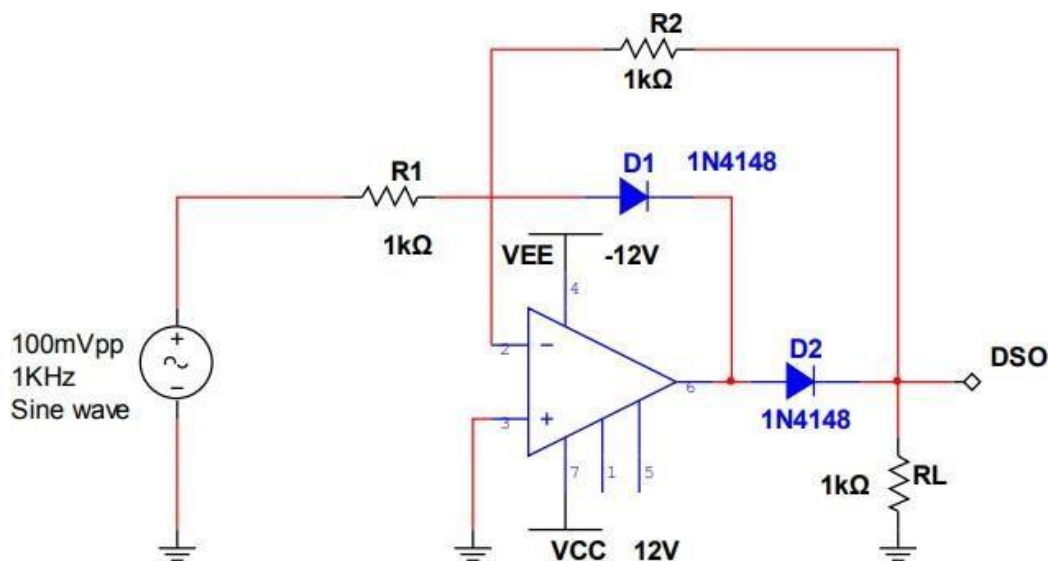
Procedure:

- 1) Connect the circuit as per the circuit diagram
- 2) Give a sinusoidal input of 100mVP, 1 kHz from a signal generator
- 3) Switch on the power supply and note down the output from CRO
- 4) Repeat the above procedure by reversing the diodes
- 5) Measure the output voltage for positive and negative half cycle
- 6) Draw input and output waveforms for each circuit

Circuit Diagram:-



(1) Non Inverting Half Wave Precision Rectifier



(2) Inverting Half Wave Precision Rectifier

Design Question:

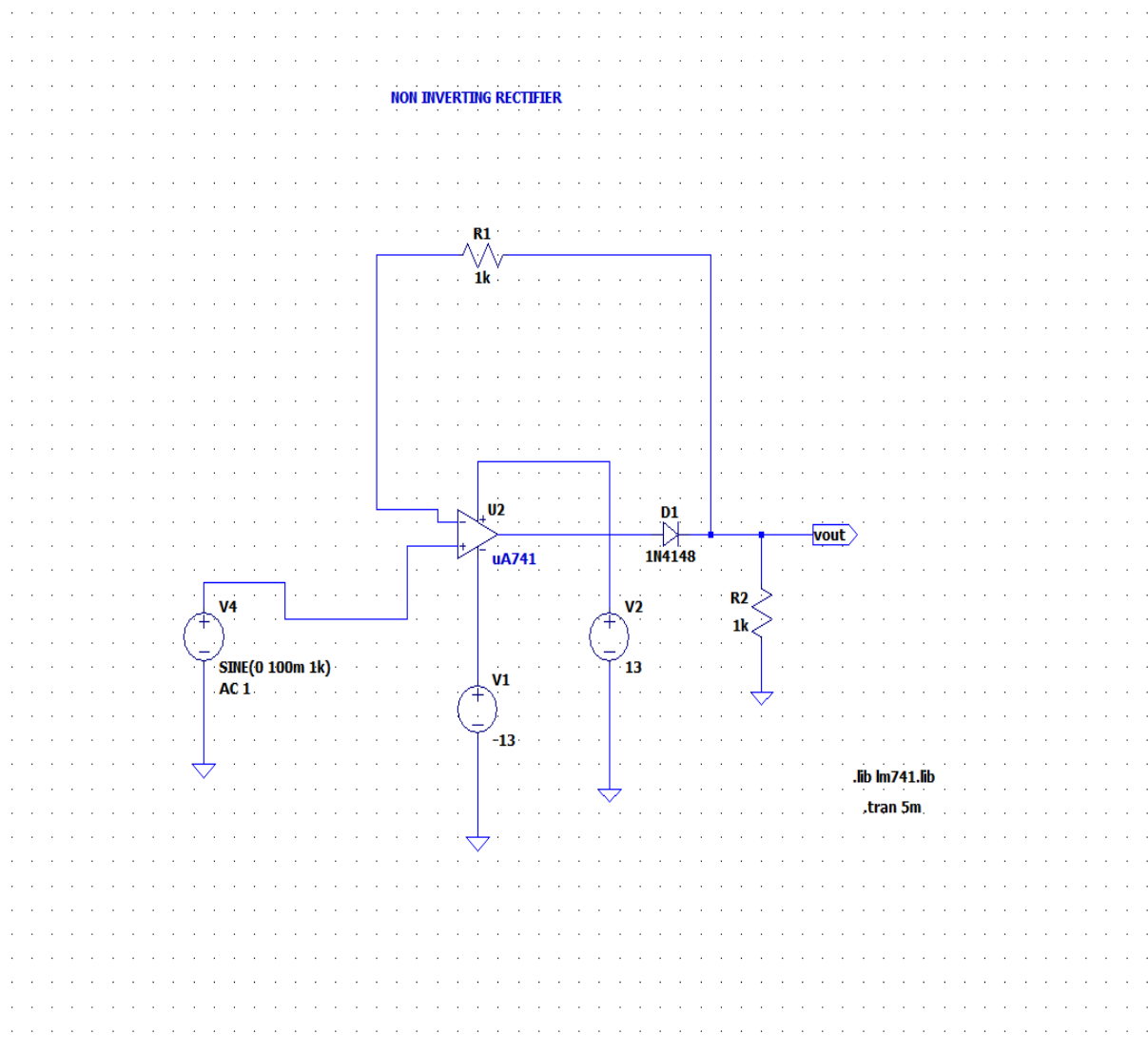
④ Design precision rectifier circuits as per lab sheet 7 and obtain the transfer curve in CRO.

Half wave rectifiers → Simulation . $V_{CC} = \pm 18V$

Full wave rectifiers → Hardware . $V_{in pp} < 1V$

* Change the direction of diodes and obtain the op waveforms .

1) NON INVERTING RECTIFIER



- Non-Inverting Half-Wave Precision Rectifier: The input signal is applied to the non-inverting (+) terminal.

Design Specifications Op-Amp:

- $\mu A741$ (High open-loop gain is critical for eliminating diode offset)
- Power Supply: $\pm 13V$ ($V_{CC} = +13V$, $V_{EE} = -13V$)
- Diode: 1N4148 Silicon switching diode. Input
- Signal: $100mV_p$ $1kHz$ Sine wave applied to Pin 3
- Load Resistance (R_L): $1k\Omega$ connected to ground to provide a discharge path

Analysis & Theory:

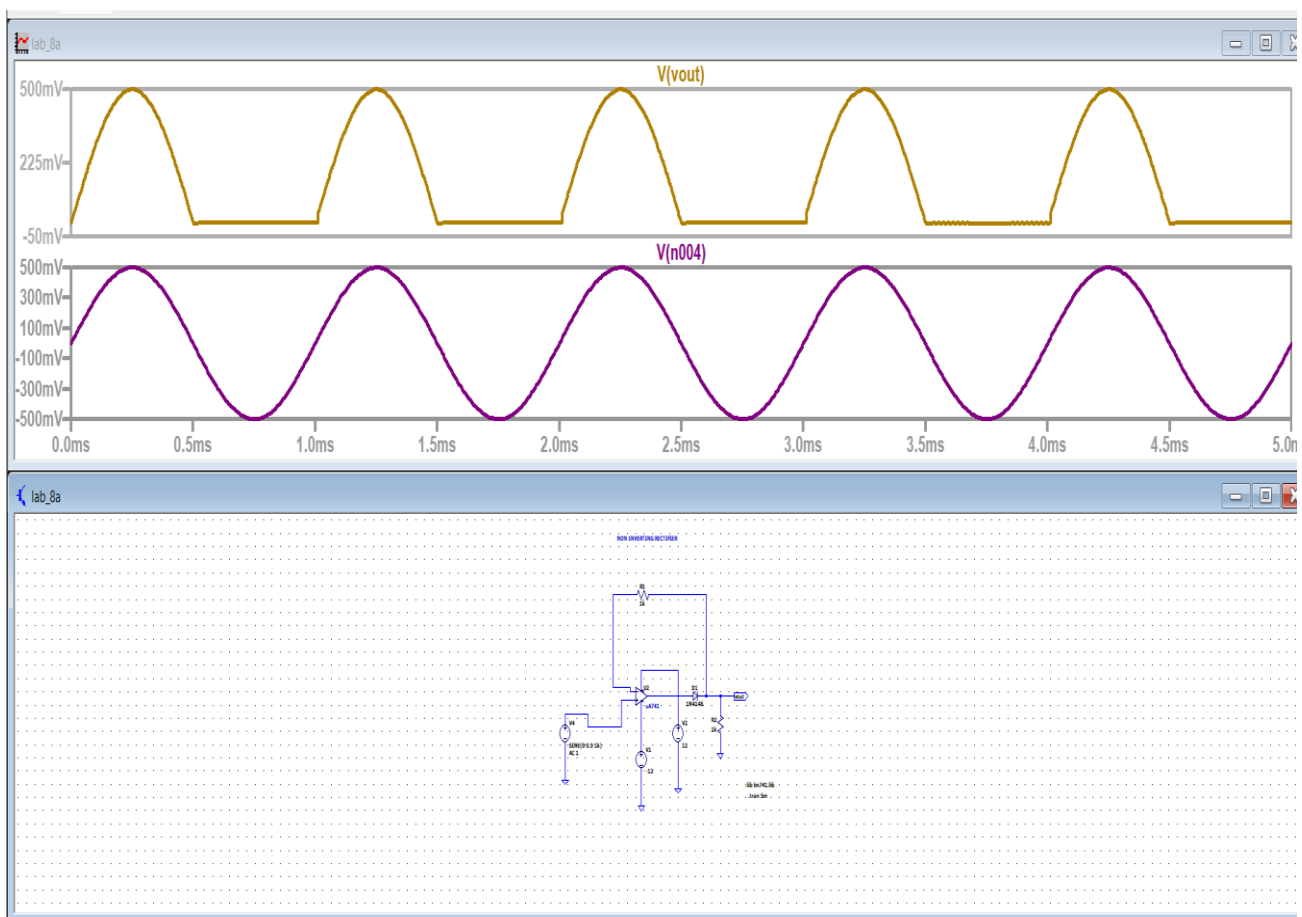
- A conventional diode requires $0.7V$ to turn on, meaning it cannot rectify small signals like $100mV$.
- Super Diode Concept: By placing the diode in the feedback loop of the op-amp, the diode's cut-in voltage is divided by the op-amp's large open-loop gain. This makes the diode act "ideally," allowing it to rectify signals in the millivolt range.
- For the Non-Inverting type: $V_{out} = V_{in}$ when $V_{in} > 0$ and $V_{out} = 0$ when $V_{in} < 0$

❖ The "Super Diode" configuration places the diode inside the feedback loop.

❖ Eliminating Cut-in Voltage: The conventional $0.7V$ drop is effectively divided by the op-amp's open-loop gain ($A_{OL} \approx 200,000$), making the diode appear nearly ideal with a $0V$ threshold.

- ❖ Positive Half-Cycle ($V_{in} > 0$): The op-amp output goes positive, forward-biasing the diode. The circuit acts as a voltage follower ($\text{Gain} = 1$). Feedback is taken from the cathode, forcing $V_{out} = V_{in}$.
- ❖ Negative Half-Cycle ($V_{in} < 0$): The op-amp output swings to the negative rail ($-13V$), reverse-biasing the diode. The feedback loop opens, and R_L pulls the output to $0V$.

Output waveforms:



- Based on the "Super Diode" theory and the specific parameters in your lab sheet, here is the detailed output analysis for both configurations when using a 0.5V Peak sine wave input

1. Non-Inverting Half-Wave Precision Rectifier IN this circuit, the op-amp compensates for the diode's 0.7V drop, allowing it to rectify the small peak 0.5V signal perfectly.

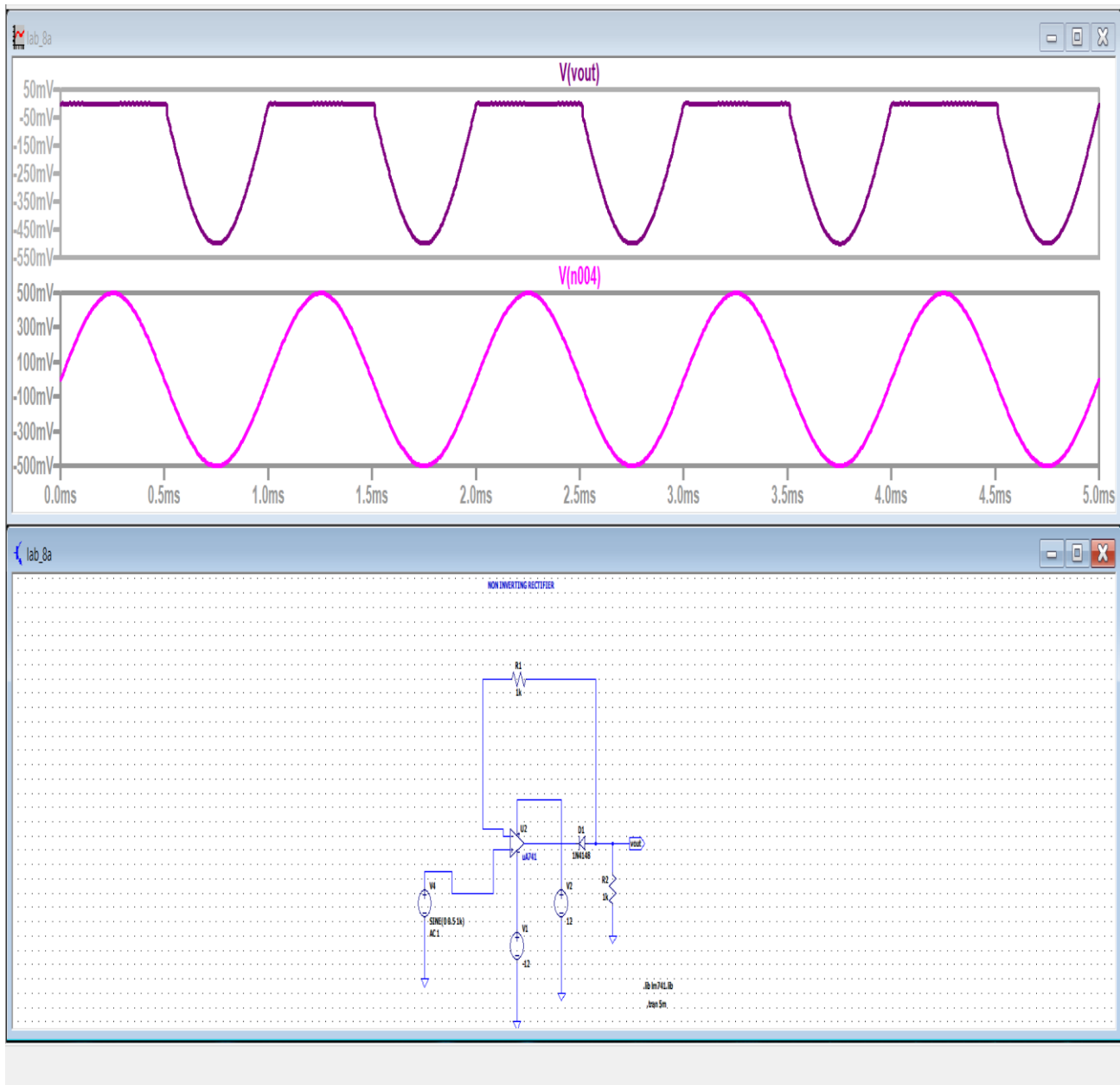
- i. Positive Half-Cycle ($V_{in} > 0$): * The op-amp output goes positive, forward-biasing the diode. The circuit acts as a voltage follower with a closed feedback loop. Output: Follows the input exactly.

- ii .Negative Half-Cycle ($V_{in} < 0$):

The op-amp output goes negative, reverse-biasing the diode. The feedback loop breaks, and the load resistor R_L pulls the output node to ground.

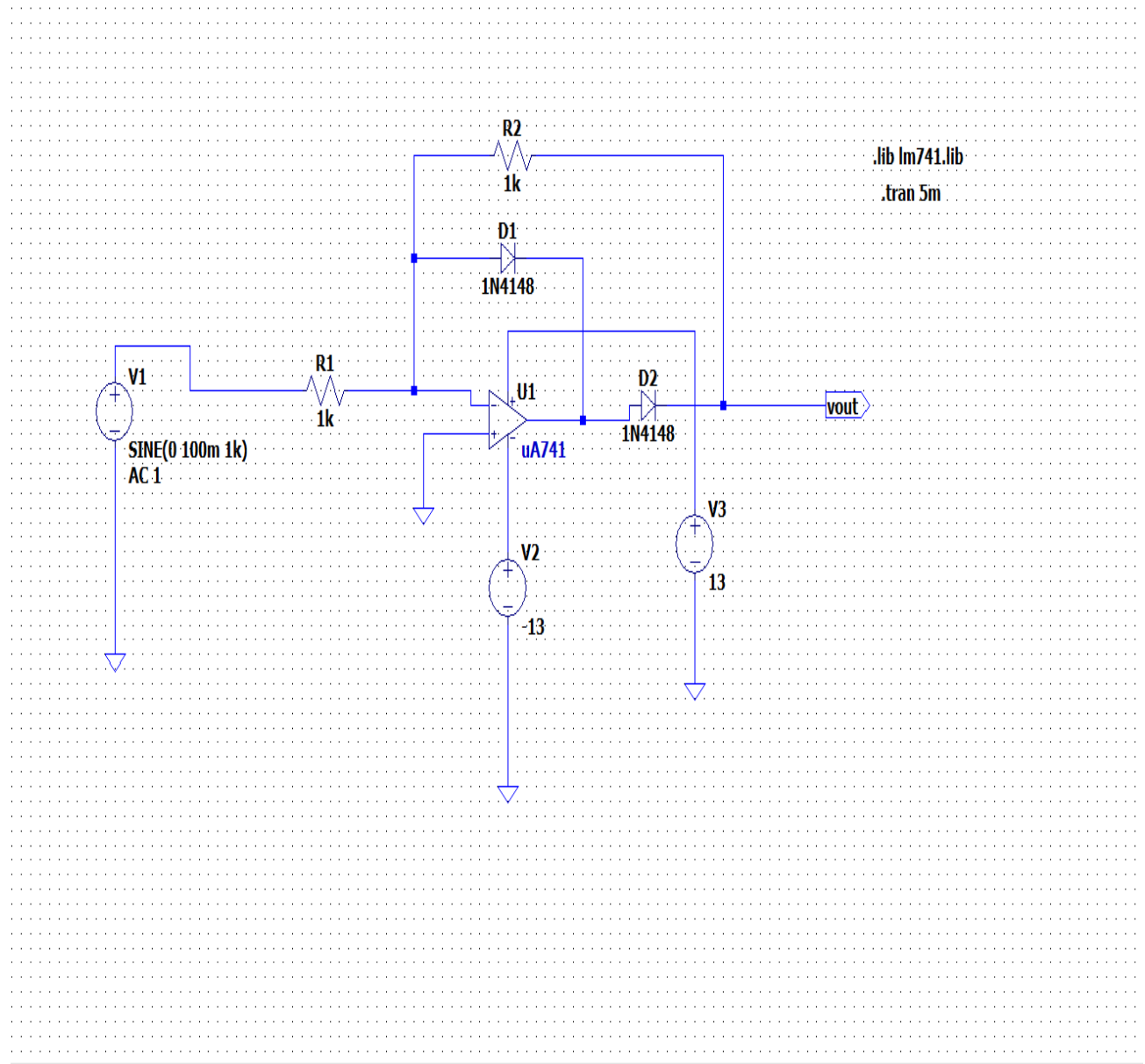
Output: A flat line at 0V .

- Changing the direction of the diodes in a precision rectifier fundamentally alters which part of the input signal is allowed to pass to the output. This is a key part of your laboratory analysis for Question 4



1. Non-Inverting Half-Wave rectifier
2. In the original design, the diode is placed to allow the positive half-cycle to pass. Original Direction (Anode to Op-Amp Output):
3. The circuit acts as a Positive Peak Rectifier. It produces an output only when $V_{in} > 0$
4. .Reversed Direction (Cathode to Op-Amp Output): The circuit becomes a Negative Peak Rectifier Behavior:
5. The diode conducts only when the op-amp output goes negative.

Inverting Half-Wave Precision Rectifier: The input signal is applied through a resistor to the inverting (-) terminal



Analysis & Theory:

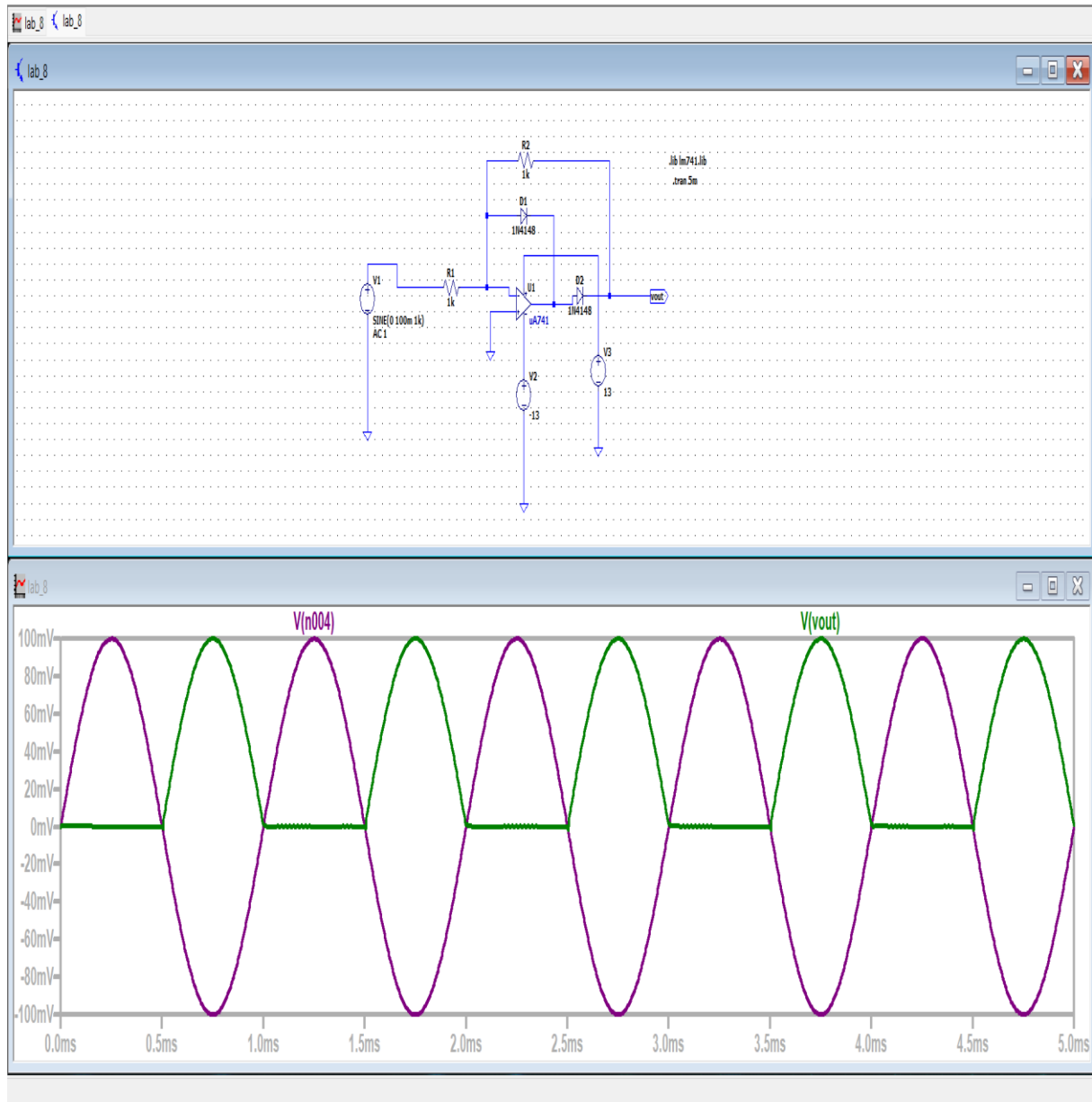
- A conventional diode requires $0.7V$ to turn on, meaning it cannot rectify small signals like $100mV$.
- Super Diode Concept: By placing the diode in the feedback loop of the op-amp, the diode's cut-in voltage is divided by the op-amp's large open-loop gain. This makes the diode act "ideally," allowing it to rectify signals in the millivolt range.
- For the Inverting type: $V_{out} = -V_{in}$ (positive half-cycle becomes a negative output) or vice-versa depending on diode direction

Theory: Preventing Saturation

- Unlike the non-inverting version, the inverting rectifier uses two diodes. D_1 provides a feedback path during the non-rectified half-cycle, which prevents the op-amp from saturating at the supply rails. This significantly improves the switching speed and frequency response of the rectifier.

Simulation Results

Positive Half-Cycle: The input is positive, the op-amp output is negative, D_2 is off, and $V_{out} = 0V$. Negative Half-Cycle: The negative input is inverted and rectified to a positive half-sine wave



Inverting Half-Wave Precision Rectifier This configuration uses two diodes (\$D_1\$ and \$D_2\$) to ensure the op-amp never goes into open-loop saturation, providing a faster and cleaner response

- positive Half-Cycle ($V_{in} > 0$):

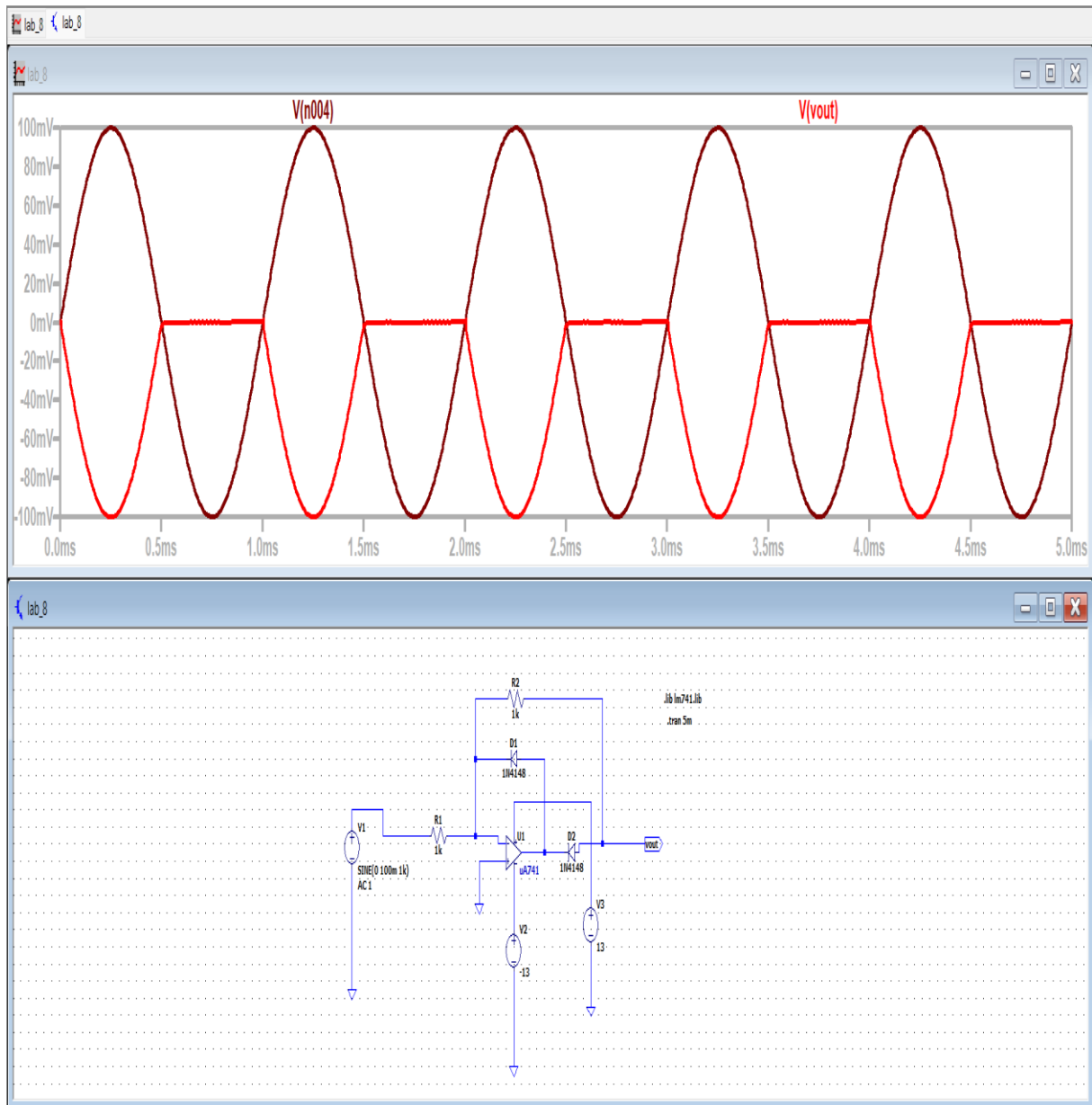
The inverting terminal tries to go positive, forcing the op-amp output negative. D_1 conducts (feedback), while D_2 is reverse-biased (blocking the output). Output: 0 V (the output is clipped during the positive input phase)

- Negative Half-Cycle ($V_{in} < 0$):

- The inverting terminal tries to go negative, forcing the op-amp output positive. D_2 conducts, allowing the signal to reach the output node, while D_1 is reverse-biased. Because it is an inverting amplifier with $R_f = R_1 = 1 \text{ k}\Omega$, the gain is -1

- The Inverting Precision Rectifier is more robust than the non-inverting version because it keeps the op-amp in a closed-loop state during both cycles.
- While it rectifies the negative half of the input, the output is always positive due to the inverting amplifier logic. This circuit is the fundamental building block for precision full-wave rectifiers

- ❖ Changing the direction of the diodes in a precision rectifier fundamentally alters which part of the input signal is allowed to pass to the output. This is a key part of your laboratory analysis for Question 4



- Inverting Half-Wave rectifier This circuit typically uses two diodes (\$D_1\$ and \$D_2\$) to prevent op-amp saturation and provide a clean rectified signal.
- Original Direction: Usually configured so that a positive input results in a rectified negative output (or vice versa). Reversed Direction (Both Diodes Flipped):
- Result: If you flip both \$D_1\$ and \$D_2\$, the circuit will rectify the opposite half-cycle.
- Example: If it was previously rectifying the positive half of the input sine wave, flipping the diodes will cause it to ignore the positive half and rectify the negative half instead.

Inferences for Inverting vs. Non-Inverting Rectifiers

▪ **Non-Inverting Half-Wave Precision Rectifier**

1. **Ideal Small-Signal Performance:** This configuration acts as a high-precision peak follower. As seen in your results, the output peak matches the input peak exactly because the Op-Amp's open-loop gain compensates for the diode's forward voltage drop.
2. **Phase Relationship:** The output is strictly in-phase with the positive half-cycle of the input.
3. **Limitation (Saturation):** During the negative half-cycle, the Op-Amp output swings to the negative supply rail (-13V) to keep the diode off. This can cause a slight delay when the input turns positive again, as the Op-Amp must "recover" from saturation.

▪ **Inverting Half-Wave Precision Rectifier**

- **Operational Stability:** This circuit is superior for high-speed applications because it uses two diodes to keep the Op-Amp in a closed-loop state at all times. One diode provides feedback during the "off" cycle, preventing the Op-Amp from hitting the saturation rails.
- **Signal Inversion:** Since it is an inverting configuration, it rectifies the negative half-cycle of the input and flips it into a positive output peak.
- **Versatility:** The gain of this circuit can be easily adjusted by changing the ratio of R_f/R_1 , whereas the non-inverting basic version is typically fixed at unity gain.

Overall Conclusion

The experiment successfully demonstrates that precision rectifiers, also known as Super Diodes, overcome the fundamental limitations of conventional diode-only rectifiers. By placing the diode within the feedback loop of a $\mu A741$ Op-Amp, the circuit effectively eliminates the 0.7V (silicon) or 0.3V (germanium) cut-in voltage requirement. This allows for the accurate rectification of millivolt-level signals, input used in this lab, with zero dead-zone and high precision.

Inference:

Precision rectifiers are critical in instrumentation and signal processing where signal amplitudes are smaller than the diode's threshold voltage. The choice between inverting and non-inverting configurations depends on whether the application prioritizes phase alignment or switching speed.